



Accuracy Test # 9

Based on NTA Abhyas (PRE-MEDICAL)

ELECTROSTATICS

1. The figure shows the electric lines of force emerging from a charged body. If the electric field at A and B are E_A and E_B respectively. If the distance between A and B is r , then



- (1) $E_A > E_B$
 (2) $E_A < E_B$
 (3) $E_A = \frac{E_B}{r}$
 (4) $E_A = \frac{E_B}{r^2}$
2. A positively charged ball hangs from a long silk thread. Electric field at a certain point (at the same horizontal level of ball) due to the ball is E . If now put a positive test charge q_0 at this point and measures $\frac{F}{q_0}$, then E

- (1) $> \frac{F}{q_0}$
 (2) $< \frac{F}{q_0}$
 (3) $= \frac{F}{q_0}$
 (4) none of these
3. Point charge $q, -q, 2Q$ and Q are placed in order at the corners A, B, C and D of a square of side $2b$. If the field at the midpoint of CD is zero, then $\frac{q}{Q}$ is

- (1) 1
 (2) 2
 (3) $\frac{2\sqrt{5}}{5}$
 (4) $\frac{5\sqrt{5}}{2}$
4. The electric potential V at any point (x, y, z) in space is given by $V = 4x^2$. The electric field E in $\left(\frac{V}{m}\right)$ at point $(1, 0, 2)$ is

- (1) 8 in $+x$ direction
 (2) 8 in $-x$ direction
 (3) 16 in $+x$ direction
 (4) 8 in $-x$ direction

5. 1000 small water drops each of radius r and charge q coalesce together to form one spherical drop. The potential of the bigger drop is larger than that of the smaller one by a factor

- (1) 1000
 (2) 100
 (3) 10
 (4) 1

6. Two spheres of radii 2 cm and 3 cm are charged to same potential. If σ_1 and σ_2 be respectively the values of surface charge density on the conductors, then the ratio $\frac{\sigma_1}{\sigma_2}$ will be

- (1) $\frac{4}{9}$
 (2) $\frac{2}{3}$
 (3) $\frac{3}{2}$
 (4) $\frac{9}{4}$

7. A Proton and an electron are released from an infinite distance apart they get attracted towards Each Other. Which of the following statement about their Kinetic energy is true?

- (1) kinetic energy of electron is more than of protons
 (2) kinetic energy of electron is less than that of Proton
 (3) kinetic energy of electron = kinetic energy of proton
 (4) none of the above is true as it depends on the distance between the particles

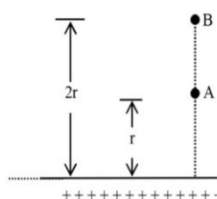
8. What is the ratio of the electrostatic potential at the corner and the centre point of a charged conducting cube? (The potential is considered 0 at infinity)

- (1) 2:1
- (2) 4:1
- (3) 1:2
- (4) 1:1

9. Two charges $4q$ and q are placed 30 cm apart. At what point the value of the electric field will be zero?

- (1) 10 cm away from q and between the charges
- (2) 20 cm away from q and between the charges
- (3) 10 cm away from q and outside the line joining the charges
- (4) 10 cm away from $4q$ and outside the line joining them

10. An infinitely long line of linear charge density λ is shown in the figure. The potential difference $V_A - V_B$ between the two points A and B is



- (1) $\frac{\lambda}{2\pi\epsilon_0} \ln 2$
- (2) $-\frac{\lambda}{2\pi\epsilon_0} \ln 2$
- (3) $\frac{\lambda}{4\pi\epsilon_0} \ln 2$
- (4) $-\frac{\lambda}{4\pi\epsilon_0} \ln 2$

11. The electrostatic potential due to an electric dipole at an equatorial point is

- (1) maximum
- (2) zero
- (3) minimum
- (4) not defined

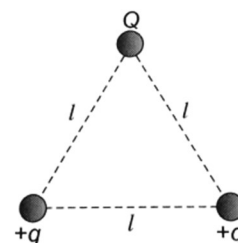
12. A wooden block performing SHM on a frictionless surface with frequency V_0 . The block carries a charge $+Q$ on its surface. If now a uniform electric field $\vec{E}$ is switched-on as shown, then the SHM of the block will be

- (1) of the same frequency and which shifted mean position
- (2) of the same frequency and with the same mean position
- (3) of change frequency and with shifted mean position
- (4) of change frequency and with the same mean position

13. On moving a charge of 20 C by 2 cm 2 J of work is done. the potential difference between the points is

- (1) 0.1 V
- (2) 8 V
- (3) 2 V
- (4) 0.5 V

14. Three Charges are placed on a vertex of an equilateral triangle as shown in figure. The value of Q for which the electrostatic potential energy of the system is zero



- (1) $-q$
- (2) $q/2$
- (3) $-2q$
- (4) $-q/2$

15. At a point in the space, the electric field points towards north. In the region surrounding this point, the rate of change of potential will be zero along

- (1) North
- (2) South
- (3) North-south
- (4) East-west

16. The surface density of charge on the surface of a charged conductor in the air is $26.5 \mu\text{Cm}^{-2}$. Then the outward force per unit area of the charged conductor is

- (1) 3.5 N/m^2
- (2) 17.7 N/m^2
- (3) 8.85 N/m^2
- (4) 39.7 N/m^2

17. The electric field of an electric dipole at a point on its axis, at a distance d from the centre of the dipole, varies as

- (1) $\frac{1}{d}$
- (2) $\frac{1}{d^2}$
- (3) $\frac{1}{d^3}$
- (4) $\frac{1}{d^{\frac{3}{2}}}$

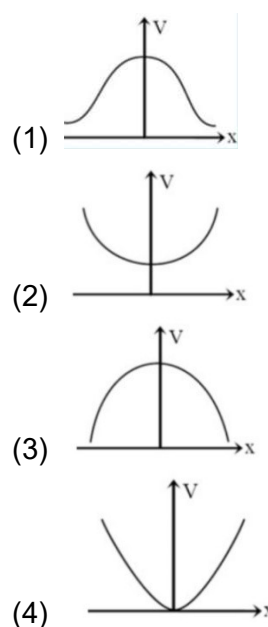
18. A charge Q is distributed over two concentric hollow spheres of radii r and R ($R > r$) such that the surface densities are equal. Find the potential at the common center.

- (1) $\frac{1}{4\pi\epsilon_0} \frac{Q(R+r)}{R^2+r^2}$
- (2) $\frac{1}{4\pi\epsilon_0} \frac{Q(R-r)}{R^2+r^2}$
- (3) $\frac{1}{4\pi\epsilon_0} \frac{Q(R+r)^2}{R^2+r^2}$
- (4) $\frac{1}{4\pi\epsilon_0} \frac{Q(R+r)^2}{R^2+2r^2}$

19. Two lithium nuclei in a lithium vapour at room temperature do not combine to form a carbon nucleus because

- (1) Carbon nucleus is an unstable particle
- (2) It is not energetically favourable
- (3) Nuclei do not come very close due to Coulombic repulsion
- (4) Lithium nucleus is more tightly bound than a carbon nucleus

20. Two identical positive charges are placed on the y -axis at $y = -a$ and $y = +a$. the variation of V (potential) along the x -axis is shown by the graph



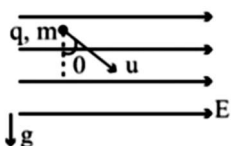
21. The magnitude of electric field intensity at point B $(2,0,0)$ due to a dipole of dipole moment $\vec{p} = \hat{i} + \sqrt{3}\hat{j}$ kept at origin is (assume that the point B is at a large distance from the dipole and $K = \frac{1}{4\pi\epsilon_0}$)

- (1) $\frac{\sqrt{13} K}{8}$
- (2) $\frac{\sqrt{13} K}{4}$
- (3) $\frac{\sqrt{7} K}{8}$
- (4) $\frac{\sqrt{7} K}{4}$

22. At any point $(x,0,0)$ the electric potential V is $\left(\frac{1000}{x} + \frac{1500}{x^2} + \frac{500}{x^3}\right)$ volt, then electric field intensity at $x=1$ m is

- (1) $5500 (\hat{i} + \hat{j}) \text{ V m}^{-1}$
- (2) $5500 \hat{i} \text{ V m}^{-1}$
- (3) $\frac{5500}{\sqrt{2}} (\hat{j} + \hat{k}) \text{ V m}^{-1}$
- (4) $\frac{5500}{\sqrt{2}} (\hat{i} + \hat{k}) \text{ V m}^{-1}$

23. A particle of mass m and charge q is thrown from a point in space where uniform gravitational field & electric field are present. The particle



- (1) May follow a straight line
- (2) May follow a circular path
- (3) May follow a parabolic path

- (1) 1 & 2 are correct
- (2) 1 & 3 are correct
- (3) 2 & 3 are correct
- (4) all these are correct

24. The electric field and electric potential at any point are E and V respectively, then

- (1) if $E=0$, V must be zero
- (2) if $V=0$, E must be zero
- (3) if $E \neq 0$, V may be zero
- (4) if $V \neq 0$, E cannot be zero

25. A charged soap bubble having surface charge density σ and radius r . if the pressure inside and outside the soap bubble is the same, then the surface tension of the solution is

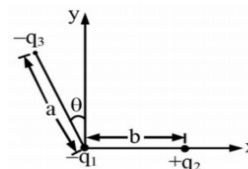
- (1) $T = \frac{\sigma^2 R}{8 \epsilon_0}$
- (2) $T = \frac{\sigma^2 R}{4 \epsilon_0}$
- (3) $T = \frac{\sigma^2 R}{2 \epsilon_0}$
- (4) $T = \frac{\sigma^2 R}{\epsilon_0}$

26. The line AA' is on charged infinite conducting plane which is perpendicular to the plane of the paper. The plane has a surface density of charge σ and B is a ball of mass m with a like charge of magnitude q . B is connected by a string from a point on the line AA' . The tangent of the angle θ formed between the line AA'' and the string is (provided, the charge q does not affect the distribution of charge on conducting plate)



- (1) $\frac{q\sigma}{2\epsilon_0 mg}$
- (2) $\frac{q\sigma}{4\pi\epsilon_0 mg}$
- (3) $\frac{q\sigma}{2\pi\epsilon_0 mg}$
- (4) $\frac{q\sigma}{\epsilon_0 mg}$

27. Three charges $-q_1$, q_2 and $-q_3$ are placed as shown in the figure. The x -component of the force on $-q_1$ is proportional to

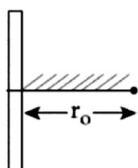


- (1) $\frac{q_2}{b^2} - \frac{q_3}{a^2} \sin \theta$
- (2) $\frac{q_2}{b^2} - \frac{q_3}{a^2} \cos \theta$

(3) $\frac{q_2}{b^2} + \frac{q_3}{a^2} \sin \theta$

(4) $\frac{q_2}{b^2} + \frac{q_3}{a^2} \cos \theta$

- 28.** A positive point charge is released from rest at a distance r_0 from a positive line charge with uniform charge density. The speed (v) of the point charge, as a function of instantaneous distance r from line charge, is proportional to



(1) $v \propto \frac{r}{r_0}$

(2) $v \propto \sqrt{\ln \left(\frac{r}{r_0} \right)}$

(3) $v \propto \ln \left(\frac{r}{r_0} \right)$

(4) $v \propto e^{+\frac{r}{r_0}}$

- 29.** An electric dipole of length 2 cm is placed with its axis making an angle 30° with a uniform electric field. If it experiences a torque of $10\sqrt{3}$ Nm, then potential energy of dipole is

- (1) - 10 J
 (2) - 20 J
 (3) - 30 J
 (4) - 40 J

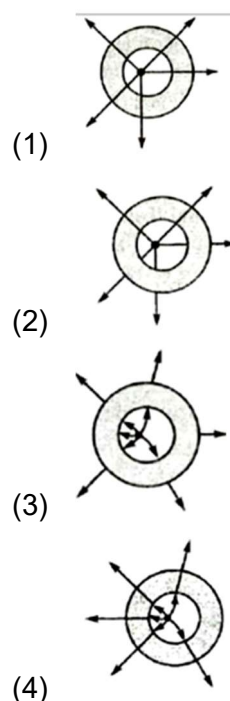
- 30.** A charged oil drop falls with terminal velocity v_0 in the absence of electric field. An electric field E keeps it stationary. The drop acquires charge $3q$, it starts moving upwards with velocity v_0 . The initial charge on the drop is

- (1) $q/2$
 (2) q
 (3) $3q/2$
 (4) $2q$

- 31.** How much positive charge should be given to the earth so as to have the same potential as that of a positively charged sphere of $1\mu\text{C}$ and radius 1 cm. (Radius of earth = 6400 km)

- (1) 600 C
 (2) 640 C
 (3) 340 C
 (4) 240 C

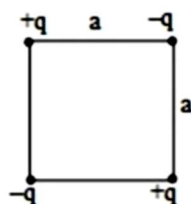
- 32.** A metallic shell has a point charge q kept inside its cavity. Which one of the following diagrams correctly represents the electric field lines?



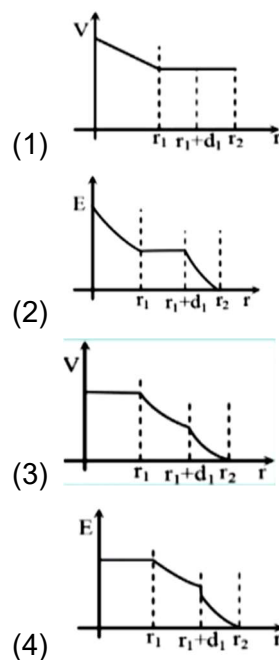
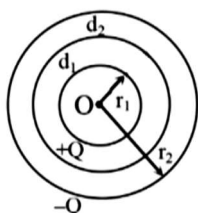
- 33.** A charge Q is divided into two charges q and $(Q - q)$. The value of q such that the force between them is maximum is

- (1) Q
 (2) $\frac{3Q}{4}$
 (3) $\frac{Q}{2}$
 (4) $\frac{Q}{3}$

34. The work required to put the four charges from infinity to the position shown here is



- (1) $\frac{-0.65q^2}{\pi\epsilon_0 a}$
 (2) $\frac{-1.0q^2}{\pi\epsilon_0 a}$
 (3) $\frac{(1-\frac{1}{2\sqrt{2}})q^2}{\pi\epsilon_0 a}$
 (4) Zero
35. A circular ring of radius R and uniform linear charge density $+\lambda$ C/m are kept in $x-y$ plane with its centre at the origin. The electric field at a point $(0,0,\frac{R}{\sqrt{2}})$ is
- (1) $\frac{\lambda}{3\sqrt{3}\epsilon_0 R}$
 (2) $\frac{2\lambda}{3\sqrt{3}R^2}$
 (3) $\frac{2}{3\sqrt{3}} \frac{\lambda}{\epsilon_0 R}$
 (4) none of these
36. Two concentric conducting spheres of radii r_1 and r_2 ($r_1 < r_2$) carry electric charges of $+Q$ and $-Q$ respectively. The region between the spheres is filled with two non-insulating layers of dielectric constant ϵ_1 and ϵ_2 and width d_1 and d_2 respectively. Variation of potential and electric field with radial distance from O is given. Select the correct one (assume V at $r_2 = 0$)



37. A charge Q placed at the centre of a metallic spherical shell with inner and outer radii R_1 and R_2 respectively. The normal component of the electric field at any point on the Gaussian surface with radius between R_1 and R_2 will be
- (1) Zero
 (2) $\frac{Q}{4\pi R_1^2}$
 (3) $\frac{Q}{4\pi R_2^2}$
 (4) $\frac{Q}{4\pi(R_1-R_2)^2}$
38. Two equally charged small metal balls placed at a fixed distance experience a force F . A similar uncharged metal ball after touching one of them is placed at the middle point between the two balls. The force experienced by this ball is
- (1) $\frac{F}{2}$
 (2) F
 (3) $2F$
 (4) $4F$

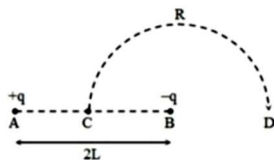
39. A particle A has a charge q and particle B has charge $4q$ with each of them having the mass m . when they are allowed to fall from rest through the same potential difference, the ratio of their speeds $v_A : v_B$ will be

- (1) 4 : 1
- (2) 1 : 4
- (3) 1 : 2
- (4) 2 : 1

40. Two non-conducting solid spheres of radii R and $2R$ having uniform volume charge densities ρ_1 and ρ_2 , respectively, touch each other. The net electric field at a distance $2R$ from the center of the smaller sphere, along the line joining the centers of the spheres is zero. The ratio $\frac{\rho_1}{\rho_2}$

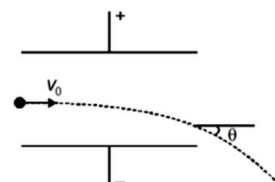
- (1) 2
- (2) 8
- (3) 4
- (4) 5

41. Charges $+q$ and $-q$ are placed at A and B, respectively, which are at a distance $2L$ apart. If C is the midpoint between A and B, then work done in moving charge Q from C to D (through semi-circular path CRD) and from C to ∞ are, respectively,



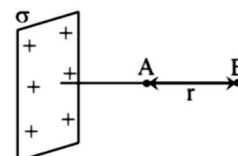
- (1) $\frac{qQ}{2\pi\epsilon_0 L}$ and infinity
- (2) $\frac{qQ}{6\pi\epsilon_0 L}$ and zero
- (3) zero, zero
- (4) $-\frac{qQ}{6\pi\epsilon_0 L}$ and zero

42. When a positive charged particle is projected with a velocity v_0 in the region between the horizontal plates of the capacitor then it leaves by making an angle θ with the horizontal, as shown. If $\theta = \theta_1$ is for proton and $\theta = \theta_2$ for α -particle then the value of $\frac{\tan \theta_1}{\tan \theta_2}$ is



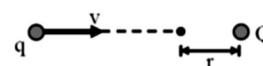
- (1) 1
- (2) $\sqrt{2}$
- (3) 2
- (4) $\frac{1}{\sqrt{2}}$

43. An infinite conducting sheet has surface charge density σ . The distance between two points is r . The potential difference ($V_A - V_B$) between these points is



- (1) $\frac{\sigma r}{2\epsilon_0}$
- (2) $\frac{\sigma r}{\epsilon_0}$
- (3) $\frac{\sigma}{\epsilon_0 r}$
- (4) $\frac{\sigma}{2\epsilon_0 r}$

44. A charged particle q is shot towards another charged particle Q (which is fixed) with a speed v . It approaches Q up to the closest distance r and then returns. If q were given a speed $2v$, the closest distance of approach would be



- (1) r
- (2) $2r$
- (3) $r/2$
- (4) $r/4$

45. A positive point charge Q is brought near an isolated uncharged metal cube

- (1) The cube becomes negatively charged
- (2) The cube becomes positively charged
- (3) The interior becomes positively charged and the surface becomes negatively charged.
- (4) The cube remains neutral.